

CURRICULUM VITAE

Dr. Kamal Kishor

Ph.D.

Department of Hematology

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ACADEMIC QUALIFICATIONS

- **PhD. (2019)** From department of hematology, All India Institute of Medical Sciences, New Delhi, India
- **Supervisor:** Dr. Renu Saxena, Professor and head, department of hematology, All India Institute of Medical Sciences, New Delhi, India
Thesis title: “Evaluate the role of tissue factor and tissue factor pathway inhibitor gene polymorphisms in the pathogenesis of Deep Vein Thrombosis in India”
- **M.Sc. (2008)** in Biochemistry from department of biochemistry, Bundelkhand university Jhansi, India
- **B.Sc. (2005)** in Zoology, Botany and Chemistry from Lucknow University, Lucknow India.

RESEARCH EXPERIENCE

- **Post doctorate research (Junior Research Fellow March 2023 to December 2023)**
Department of Hematology, All India Institute of Medical Sciences, New Delhi, India.
Project 1 title: “To assess the utility of artificial intelligence and machine learning to predict cancer associated venous thromboembolism and compare it with existing biomarkers and prediction scores”.
Investigator: Dr. Mukul Aggarwal
- **Post doctorate research (Senior Research Fellow May 2019 to March 2023) in**
Department of Hematology, All India Institute of Medical Sciences, New Delhi, India.
Project 2 title: “Mitigation of clinical phenotype by activated platelets in severe haemophilia”.

Investigator: Dr. M. Mahapatra

Project 3 title “Proposal for engagement of All India Institute of Medical Sciences to serve as quality controls (QC) lab for comprehensive national nutrition survey”.

Investigator: Dr. R Lakshmy

- **Senior Research Fellow (May 2018 to April 2019)** in Department of Hematology, All India Institute of Medical Sciences, New Delhi, India.

Project title: “Diagnostic efficacy of digital hemoglobinometer (TrueHB), HemoCue and non-invasive spectroscopic devices for screening patients for anemia in the field setting”.

Investigator: Dr. Renu Saxena

- **Research Scholar (August 2008 – Feb 2011)** in Department of Hematology, All India Institute of Medical Sciences, New Delhi, India.

Project title: “Role of nitric oxide in the development of stroke in young Indian”

Investigator: Dr. Renu Saxena

TECHNICAL PROFICIENCY

- **Molecular Biology:** Isolation and estimation of genomic DNA/RNA, cDNA synthesis, PCR, allele specific PCR, multiplex PCR, nested PCR, RT-PCR, quantitative real-time PCR, primer designing, restriction digestion, VNTRs, Conformation Sensitive Gel Electrophoresis (CSGE), Single Stranded Conformation Polymorphism (SSCP), Silver staining, Agarose gel electrophoresis, Polyacrylamide gel electrophoresis, Image acquisition and Analysis.
- **Flow cytometry:** Working experience on BD FACS Canto II, platelets activation markers such as P-selectin, GP-IIb/IIIa and Phosphatidyl Serine (PS) expressions by using CD61, CD62P, PAC-1 and Annexin V binding. Platelets’ Mitochondrial Membrane Potential by using JC-1 dye binding. Stem cells characterization, immunophenotyping of leukemia, PNH, BSS and Glanzmann thrombasthenia.
- **Biochemistry:** ELISA, Western Blotting.
- **Proteomics:** Isolation and estimation of proteins, PAGE-SDS & Native.
- **Hematological diagnosis:** Carrier detection and prenatal diagnosis of Hemophilia, molecular diagnosis of thrombophilia, mutations detection in thalassemia. JAK 2, BCR-ABL and PML-RARA for leukemia. Coagulation factors activity Assay, platelet function disorder like Bernard-Soulier syndrome, Von Willebrand disease and

Glanzmann thrombasthenia by platelet aggregation study and diagnosis of thalassemia by HPLC.

- **Bioinformatics and computer skills:** Knowledge of computational and statistical analysis of biological systems by using software viz. STATA and Graph Pad, BLAST, Primer design and RFLP design using NCBI and other web based bioinformatics tools.
- **Reference Management:** Mendeley and EndNote

AWARDS AND HONORS

- Awarded international travel grant from CSIR to attend the ISTH conference which was held in London, England UK during July 2022.
- Awarded AIIMS Endowment partial fund to attend the ISTH conference which was held in London, England UK during July 2022.
- Awarded Reach the world travel grant from ISTH to attend the conference which was held in Philadelphia, USA, in July 2021.
- Awarded Senior research fellow from Department of Biotechnology (DBT) from Nov. 2019 to till date.
- Awarded Senior research fellow from UNICEF May 2019 to Oct. 2019.
- Awarded Senior research fellow from Ministry of Health and family Welfare from May 2018 to April 2019.
- Awarded Reach the world travel grant from ISTH to attend the conference which was held in Melbourne, Australia in July 2019.
- Awarded AIIMS Endowment partial fund to attend the XXVII Congress of the International Society on Thrombosis and Haemostasis which was held in Melbourne, Australia from 6th July to 10th July 2019.
- Awarded international travel grant from DST to attend 63rd Annual SSC Meeting of the ISTH which was held in Berlin, Germany during July 2017.
- Awarded Dr. K. C. Das Memorial Award from ISHBT for best published article in the field of basic research in laboratory hematology for the year 2018.
- Awarded Student scholar ship from ISHBT for attending 59th Annual meeting of ISHBT which was held in Kochi during 25-28, October 2018.
- Awarded Institute Fellowship grant (AIIMS) from 1st Feb 2012 to 16th Feb 2017.
- Awarded best poster award on topic OBESITY in KEPLER 07 held in Bundelkhand University Jhansi Indian 2007.

MEMBERSHIP OF PROFESSIONAL BODIES

- Indian Society of Hematology & Blood Transfusion (ISHBT)–**LIFETIME MEMBER**
- International society of Thrombosis and Haemostasis (ISTH) - **ANNUAL MEMBER**

PUBLICATIONS/ABSTRACTS

Publications in various national/International journals:

1. Samarjit Maharana, Chitrani Laha Roy, **Kamal Kishor**, Ravi Ranjan, Firdos Ahmad, Manoranjan Mahapatra, Renu Saxena, Meganathan Kannan. Depolarized mitochondrial membrane potential and elevated calcium in platelets of Sickle Cell Disease. IJHB-D-22-00732R1 accepted.
2. **Kishor K**, Sharma A, Singh K, Ranjan R, Pandey H, Kumar R, Kamal VK, Mishra P, Mahapatra M, Saxena R. Influence of tissue factor polymorphisms (603A>G and 5466A>G) on plasma tissue factor levels and their impact on deep vein thrombosis risk in young Indian population. J Thromb Thrombolysis. 2018 Jul;46(1):88-94.
3. **Kishor K**, Sharma A, Singh K, Ranjan R, Pandey H, Kumar R, Mishra P, Saxena R. Association of genetic polymorphisms with plasma TFPI level: Boon or curse for DVT patients - Study from India. Blood Cells Mol Dis. 2017 Jul;66:31-36.
4. Pandey H, Singh K, Ranjan R, Pandey S, Sharma A, **Kishor K**, Seth T, M Mahapatra, Saxena R, Clinical variability and molecular characterization of Hbs/ G γ (A γ δ β)0-thal and Hbs/HPFH in Indian sickle cell disease patients: AIIMS experience. Hematology, 2019;24(1):349-352.
5. Sharma A, Singh K, Biswas A, Ranjan R, **Kishor K**, Pandey H, Kumar R, Mahapatra M, Oldenburg J, Saxena R. Impact of interleukin 6 promoter polymorphisms (-174 G > C, -572 G > C and -597 G > A) on plasma IL-6 levels and their influence on the development of DVT: a study from India. Hematology. 2018 Dec;23(10):833-838.
6. Pandey H, Ranjan R, Singh K, Sharma A, **Kishor K**, Seth T, Saxena R. Contrasting co-inheritance of alpha and beta mutations in delta beta thalassemia and hereditary persistence of fetal hemoglobin: a study from India. Hematology. 2018 Oct;23(9):692-696
7. Sharma A, Singh K, Biswas A, Ranjan R, **Kishor K**, Kumar R, Pandey H, Kamal VK, Saxena R. Evaluation of role of FV, FVIII and APLAs in the pathogenesis of APCR in FV Leiden negative DVT patients: a study in India. J Thromb Thrombolysis. 2017 Feb;43(2):217-223.
8. Sharma A, Bhakuni T, Biswas A, Ranjan R, Kumar R, **Kishor K**, Mahapatra M, Jairajpuri MA, Saxena R. Prevalence of Factor V Genetic Variants Associated With Indian APCR Contributing to Thrombotic Risk. Clin Appl Thromb Hemost. 2017 Sep;23(6):596-600.
9. Sharma A, Singh N, Mahapatra M, Ranjan R, **Kishor K**, Saxena R. Factors contributing to APC-resistance in women with recurrent spontaneous miscarriages: Indian perspective. Blood Cells Mol Dis. 2015 Oct;55(3):213-215.
10. Sharma A, Bhakuni T, Ranjan R, Kumar R, **Kishor K**, Kamal VK, Mahapatra M, Jairajpuri MA, Saxena R. Polymorphisms in factor V and antithrombin III gene in recurrent pregnancy loss: a case-control study in Indian population. J Thromb Thrombolysis. 2015 May;39(4):481-488.

11. Pandey SK, Meena A, **Kishor K**, Mishra RM, Pandey S, Saxena R Prevalence of Factor V Leiden G1691A, MTHFR C677T, and Prothrombin G20210A Among Asian Indian Sick Cell Patients Clinical and Applied Thrombosis Hemostasis 2012 18(3) 320-323.
12. Chopra A, Kumar R, **Kishore K**, Tandon N, Yusuf T, Kumar S, Mishra P and Saxena R Effect of glucocorticoids on von Willebrand factor levels and its correlation with von Willebrand factor gene promoter polymorphism Blood Coagulation and Fibrinolysis 2012, 23:514–519.
13. Ahmad F, Kannan M, **Kishor K** and Saxena R Coinheritance of Severe von Willebrand Disease with Glanzmann Thrombasthenia Clinical and Applied Thrombosis Hemostasis 2010 16(5) 529-532.

Abstracts in various International/national conferences:

1. **Kamal Kishor**¹, Amit Sharma¹, Ravi Ranjan¹, Ravi Kumar¹, Jasmita Dass, Renu Saxena², Manoranjan Mahapatra¹ **Role of tissue factor intronic polymorphism 5466A>G in Indian thrombophilic patients. ISTH 2023 (Accepted)**
2. **Kamal Kishor**¹, Amit Sharma¹, Ravi Ranjan¹, Ravi Kumar¹, Jasmita Dass, Renu Saxena², Manoranjan Mahapatra¹ Role of Tissue factor gene polymorphism 603A>G with Recurrent pregnancy loss in India. BBIF 2023 India
3. **Kamal Kishor**, Amit Sharma, Samarjit Maharana, Chitrani Laha Roy, Ravi Ranjan, Ravi Kumar, Manoranjan Mahapatra, Renu Saxena. Impact of Tissue factor promoter gene polymorphism with Recurrent pregnancy loss in India. ISTH 2022
4. Chitrani Laha Roy, Samarjit Maharana, **Kamal Kishor**, Ravi Ranjan, Manoranjan Mahapatra, Renu Saxena, Meganathan Kannan. Role of activated platelets in compensation of bleeding severity in Hemophilia A. ISTH 2022
5. Amit Sharma, **Kamal Kishor**, Arijit Biswas, Ravi Kumar, Johannes Oldenburg, Renu Saxena, Manoranjan Mahapatra. Study the of endothelial protein C receptor gene polymorphisms in Indian DVT, ISTH 2022
6. Ravi Ranjan, **Kamal Kishor**, Amit Sharma, Hareram Pandey, Jasmita Das, Mukul Aggrawal, Seema Tyagi, Manoranjan Mahapatra Real Time PCR based detection of Venous Thromboembolism associated Genetic Variations: Factor V Leiden and MTHFR C677T Polymorphism. Hematocon 2022
7. **Kamal Kishor**, Amit Sharma, Samarjit Maharana, Ravi Kumar, Ravi Ranjan, Seema Tyagi, Renu Saxena, Manoranjan Mahapatra. Impact of Tissue Factor Pathway Inhibitor gene polymorphisms (33T/C and 264V/M) on plasma TFPI levels and their influence on risk of Recurrent Pregnancy Loss in India. ISTH 2021(PB 1302)
8. **Kamal Kishor**, Amit Sharma, Ravi Kumar, Samarjit Maharana, Ravi Ranjan, Seema Tyagi, Renu Saxena, Manoranjan Mahapatra. Impact of 399C/T polymorphism of Tissue Factor Pathway Inhibitor on risk of Recurrent Pregnancy Loss in India ISTH 2020.
9. Amit Sharma, **Kamal Kishor**, Arijit Biswas, Ravi Ranjan, Seema Tyagi, Renu Saxena, Oldenburg J, Manoranjan Mahapatra Role of Endothelial Protein C Receptor gene polymorphisms (4678C>G and 6936A>G) in young Indian DVT patients ISTH 2020.

10. Samarjit Maharana, **Kamal Kishor**, Amit Sharma, Ravi Ranjan, Manoranjan Mahapatra, Renu Saxena, Meganathan Kannan Alternate pathway of platelet destruction in Sick Cell Disease. EHA 2020 [PB2360]
11. **Kamal Kishor**, Amit Sharma, Ravi Ranjan, Ravi Kumar, Hareram Pandey, Manoranjan Mahapatra, Renu Saxena. Influence of TFPI gene polymorphisms (33T/C and 536C/T) on DVT in young Indians. ISHBT 2019.
12. Panday H, Ranjan R, Singh K, Sharma A, **Kishor K**, M Mahapatra, Saxena R, Frequency of alpha gene number and HMOX1 polymorphism in Indian sickle cell disease patients. ISHBT 2019.
13. **Kamal Kishor**, Amit Sharma, Ravi Kumar, Ravi Ranjan, Hareram Pandey, Samarjit Maharana, Manoranjan Mahapatra, Renu Saxena, TFPI gene polymorphisms (399C>T, 287T/C): Do they predict high DVT risk in young Indians? ISTH 2019
14. Amit Sharma, Arijit Biswas, Ravi Ranjan, **Kamal Kishor**, Hareram Pandey, Ravi Kumar, M. Mahapatra, J. Oldenburg, Renu Saxena, Laboratory phenotype and molecular characterization of inherited coagulation Factor VII (FVII) deficiency patients from India ISTH 2019
15. **Kamal Kishor**, Amit Sharma, Hareram Pandey, Ankita Chaudhary, Ravi Ranjan, Ravi Kumar, Manoranjan Mahapatra, Renu Saxena, influence Of Tissue Factor Promoter Polymorphism on Risk of Deep Vein Thrombosis. ISHBT 2018.
16. Amit Sharma, Ravi Ranjan, **Kamal Kishor**, Hareram Pandey, Ravi Kumar, Kanwaljeet Singh, Manoranjan Mahapatra, Renu Saxena. HR2 Haplotype and Arg485Lys Polymorphisms: Association with FV levels, APC Ratio and their role in the development of DVT. ISHBT 2018.
17. Panday H, Ranjan R, Sharma A, **Kishor K**, Seth T, M Mahapatra, Saxena R, Molecular and hematological characterization of $\delta\beta$ thalassemia and HPFH with inherited cofactors. ISHBT 2018.
18. **K. Kishor**, A. Sharma, R. Ranjan, H. Pandey, P. Mishra, R. Saxena Role of tissue factor 603A>G and 5466A>G polymorphisms in the development of DVT in young Indians. 63rd Annual SSC Meeting of the ISTH 2017.
19. R. Saxena, **K. Kishor**, A. Sharma, R. Ranjan, P. Mishra, M. Mahapatra. Tissue Factor Pathway Inhibitor polymorphism 399C>T predisposes to Idiopathic DVT in young Indians. 63rd Annual SSC Meeting of the ISTH 2017.
20. A. Sharma, T. Bhakuni, A. Biswas, R. Ranjan, **K. Kishor**, J. Oldenburg, R. Saxena. The association of Arg485Lys polymorphism with FV levels, APC ratio and its role in DVT: A study from India. 63rd Annual SSC Meeting of the ISTH 2017.
21. **Kamal Kishor**, Ravi Kumar, Amit Sharma, Hare Ram Pandey, Ravi Ranjan, Pravas Mishra, Renu Saxena Association of tissue factor pathway inhibitor gene polymorphisms 399C/T and 33T/C with deep vein thrombosis EMLTD-ICT Congress 2016.

22. Sharma A, Bhakuni T, Biswas A, Ranjan R, **Kishor K** Mahapatra M, Jairajpuri M.A, Saxena R. “Association of elevated FVIII levels and APLAs with APCR^{-FV} Leiden patients with deep vein thrombosis” Abstract accepted for poster presentation in **62nd SSC meeting, 2016** at Montpellier, France.
23. Sharma A, Bhakuni T, Ranjan R, Kumar R, **Kishor K**, Mahapatra M, Jairajpuri M.A, Saxena R. “Factors associated with activated protein C resistance in Indian patients with RPL” Abstract accepted for poster presentation in **International Conference of Hematology, 2016** held at Glassglow, U. K.
24. Mishra P, **Kishor K**, Seth T, Mahapatra M and Saxena R Inadequate anticoagulation is the only risk factor for development of post thrombotic syndrome in unprovoked deep vein thrombosis ISTH 13 (Suppl. 2) (2015) 1–997.
25. Sharma A, Bhakuni T, Ranjan R, Kumar R, **Kishor K**, Mahapatra M, Jairajpuri M.A, Saxena R. “Causative factors of activated protein C resistance in Indian DVT patients.” Oral presentation in **ISHBT 2015** at Calcutta, India.
26. **Kamal Kishor**, Amit Sharma, Ravi Kumar, Hareram Pandey, Ravi Ranjan, Renu Saxena Role of promoter polymorphisms of Interleukin 6 (IL-6) gene in Indian Patients with DVT. Thrombosis Research May 2014 Volume 133, Supplement 3, Page S61.
27. Amit Sharma, Teena Bhakuni, Ravi Ranjan, Suhail Akhter, Ravi Kumar, **Kamal Kishor**, M. Mahapatra, M. A. Jairajpuri, Renu Saxena Genetic basis of Antithrombin deficiency in Indian patients with Deep Vein Thrombosis. Thrombosis Research May 2014 Volume 133, Supplement 3, Page S61.
28. Amit Sharma, Ravi Ranjan, Suhail Akhter, **Kamal Kishor**, Hareram Pandey, Renu Saxena, A-3 Haplotype of EPCR gene: Is it a risk factor for recurrent miscarriage in Indian population ? (ISHBT 2012).
29. Amit Sharma, Arijit Biswas, Ravi Ranjan, Suhail Akhter, Ravi Kumar, **Kamal Kishor**, Renu Saxena. Role of A3 haplotype of EPCR in the development of deep vein thrombosis in Asian Indian Thrombosis Research vol.130 sup.1, Page S148-149, October 2012.
30. Mohd. Suhail Akhter, Arijit Biswas, Ravi Ranjan, Amit Sharma, **Kamal Kishor** Renu Saxena The association of functional polymorphisms of IL-6 gene promoter with ischemic stroke Abstract submitted to 22nd International Congress on Thrombosis 2012.

WORKSHOPS/CONFERENCES ORAL AND POSTER PRESENTATION

- Attended Next Generation Sequencing workshop which was held in ILBS, Delhi , during 27th -28th July 2022.
- Presented a poster at ISTH 2022 conference which was held in London, England UK, during July 2022.
- Virtually presented a poster at ISTH 2021 conference which was held in Philadelphia USA during July 17-21, 2021.

- Attended virtual “India International Science Festival” (IISF) 2020 as a jury member during Dec. 22-24, 2020.
- Virtually presented a poster at ISTH 2020 virtual congress, during July 12-14, 2020.
- Oral presentation at 60th Annual conference of Indian Society of Hematology and Blood Transfusion (HEMATOCON 2019) which was held in Delhi, India during Nov.07-10, 2019.
- Presented a poster at ISTH 2019 conference which was held in Melbourne, Australia, during July 6-10, 2019.
- Presented a poster at 59th Annual meeting of ISHBT which was held at Kochi Kerala during October 25-28, 2018.
- Attended workshop on Cytogenetics at 59th Annual meeting of ISHBT which was held at Kochi, Kerala in India during October 25-28, 2018.
- Attended National Hematology Update-XI April 14th -15th 2018, All India Institute of Medical Sciences, New Delhi.
- Participated as a resource person in the pre-conference workshop on Hemophilia, National Hematology Update-XI April 13th 2018, All India Institute of Medical Sciences, New Delhi.
- Presented a poster at the 63rd Annual SSC Meeting of the ISTH which was held in Berlin, Germany, during July 8-13, 2017.
- Attended National Hematology Update-X April 2nd- 3rd 2015, All India Institute of Medical Sciences, New Delhi
- Attended SYSCON-2013 on Interfacing Basic and Translational Research 2013 All India Institute of Medical Sciences, New Delhi.
- Attended National Hematology Updates-IX 2012 at New Delhi, India, Organized by Dept. of Hematology, AIIMS.
- Attended National Hematology Updates-VIII Feb. 26th -27th 2011 at New Delhi, India, Organized by Dept. of Hematology, AIIMS.
- Attended SYS Lecture Series on “Stem cell therapies for autoimmune diseases” Feb. 5th 2011, All India Institute of Medical Sciences, New Delhi.
- Attended National Hematology Updates 2010 at New Delhi, India, Organized by Dept. of Hematology, AIIMS.
- Attended National Hematology Updates 2009 at New Delhi, India, Organized by Dept. of Hematology, AIIMS.
- Attended Golden Jubilee conference of Indian Society of Hematology & Transfusion Medicine Nov. 19th -22nd 2009, All India Institute of Medical Sciences, New Delhi.
- Attended National Hematology Update-VII Feb. 23rd -24th 2008 at All India Institute of Medical Sciences, New Delhi.
- Attended CME on Genetics sponsored by DBT, Govt. of India April 13th 2008, All India Institute of Medical Sciences, New Delhi.
- Attended SHC India-AIIMS Haemophilia conference under the aegis of Delhi Society of Haematology, March 13th -14th 2008, All India Institute of Medical Sciences, New Delhi.
- Presented a poster on topic OBESITY in KEPLER -07 held in B.U. Jhansi 2007.

REFERENCES

Dr. Renu Saxena
(Former Head of Dept AIIMS, New Delhi)
Director Dept of Path and Lab medicine and
Head hematopathology
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Email: renusaxena@outlook.com

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Declaration: I hereby declare that the particulars furnished herein by me are true to the best of my knowledge and belief.

Place:

Date:

(Dr. Kamal Kishor)